BIOINFORMATICS DATA ENGINEERING / PHD INTERVIEW SUMMARY

CancerOmicsLake

An auditable platform for studying the sensitivity of cancer-gene rankings

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Code and project repository

1. Research question and engineering contribution

How much can a cancer-gene ranking change when its reference population, expression processing, tie handling, or numerical environment changes, even when the workflow executes reproducibly? CancerOmicsLake addresses this question by connecting public data acquisition, harmonization, analytical tables and recorded experiments. Its strongest contribution is the implementation and evaluation of an inspectable research workflow, including evidence of its limitations.

Data and scope

The project integrates open-access GDC/TCGA metadata, gene expression, clinical and biospecimen fields, somatic mutation summaries, GTEx normal-tissue expression, and publicly released recount3 processed expression [1]. The cancer cohorts are breast invasive carcinoma (BRCA), lung adenocarcinoma (LUAD) and colon adenocarcinoma (COAD). Normal references are breast, lung, transverse colon and sigmoid colon. The frozen primary expression inventory contains **36,456,660 TCGA** and **11,240,000 GTEx** measurement rows; these are gene–specimen measurements, not counts of patients or sequencing files.

GDC / GTEx / recount3 / annotations

↓ configured acquisition, retries, checksums and source metadata

Bronze: immutable inputs ⟶ Silver: typed, harmonized Parquet ⟶ Gold: analytical marts

↓ DuckDB / dbt tests / quality gates / frozen experiment manifests

SQL, API, dashboard, graph exports and claim-to-source evidence

Implemented capabilities

Python/Polars ingestion and transformation support configurable cohorts and bounded downloads; DuckDB and dbt provide queryable dimensions and facts. Orchestration, Docker and CI support repeatable runs. Streamlit, FastAPI and Neo4j/Graphify exports expose the results. Validation checks cover access, checksums, types, duplicate keys, gene/sample joins and relationships.

The analytics include expression and reference comparisons, consequence-stratified mutation frequencies, candidate prioritization, pathway overrepresentation with Benjamini–Hochberg correction within cancer/candidate set, and graph degree summaries. These provide exploratory context; pathway membership, mutation frequency and graph connectivity do not by themselves validate a candidate.

Research process and contribution

Work progressed from infrastructure to reference sensitivity, exact specimen matching, tie diagnostics, published baselines and numerical replay. Repairs addressed duplicate measurements, Ensembl version normalization while preserving `_PAR_Y` loci, and donor/case leakage across partitions. Plans and input hashes preserve earlier evidence when a later study changes the design. Individual manifests and per-gene intermediates stay outside the aggregate brief.

Finding. A functioning data platform makes the assumptions and failure modes of candidate rankings inspectable. The studies below assess what that infrastructure can establish scientifically.

2. Matched specimens: broad agreement, unstable leading lists

Study design

Matched-processing v1 validated **617 exact specimens**: 417 TCGA tumors and 200 GTEx normals. Matching checked GDC sample identity or GTEx SAMPID, case/donor, tissue, QC and count-column mapping. Ambiguities were excluded and corrected measurement keys were required to be unique. The three primary comparisons use 150 normals; the remaining 50 sigmoid-colon specimens are retained separately. COAD uses transverse colon for the primary comparison. Both processing pipelines use identical specimens.

Native TPM and recount3 AUC-scaled coverage are kept in their respective units. V1 compared within-specimen gene ranks, detection and top-expressed genes, then compared tumor–normal effects. V2, designed after inspecting v1, examined ties, detection, annotation and workflow provenance. It is an exploratory follow-up.

For each gene and pipeline, the signed rank-biserial effect is

$$r_g = \frac{1}{n_T n_N} \sum_{i=1}^{n_T} \sum_{j=1}^{n_N} \operatorname{sign}(x_{gi}^T - x_{gj}^N).$$

This measures relative ordering within a pipeline. It does not make expression units comparable or provide a differential-expression significance test.

Matched v2: complete shared-gene universe (55,660 genes per cancer)

Cancer	Tumors	GTEx	τ_b	Direction agreement	Eligible genes	Mean J_{50}
BRCA	162	50	0.559	89.85%	23,247	0.0046
COAD	119	50	0.556	90.53%	22,007	0.0026
LUAD	136	50	0.571	90.21%	23,732	0.0058

τ_b: signed-effect Kendall correlation. Direction agreement uses genes with $|r_g| \geq 0.2$ in both pipelines; its denominator is shown. J_{50}: Jaccard overlap of absolute-effect top-50 lists after seeded boundary-tie resolution. Specimens are not the denominator for these gene-level metrics.

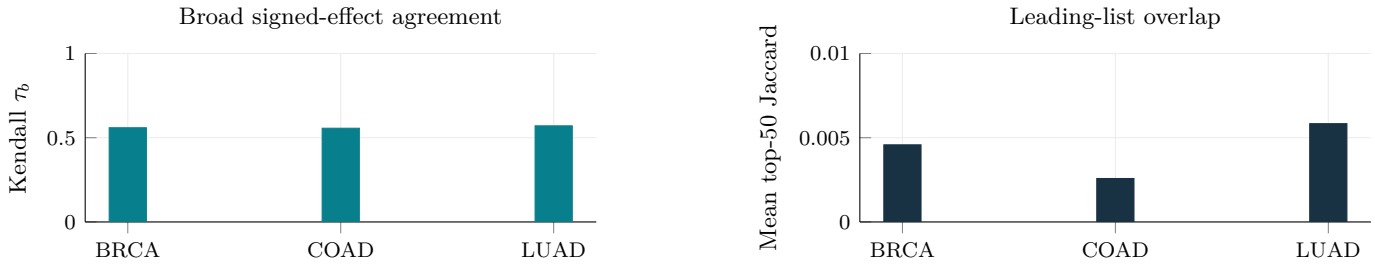


Figure: the panels use different vertical scales and different statistics; their bar heights should not be compared directly. Values come from the same matched comparisons in the table.

V2 evaluated all shared genes and a fixed informative subset detected (> 0) in at least 20% of both groups in both pipelines. It used 1,000 seeded tie resolutions and 200 paired case/donor bootstrap replicates, with 100 tie resolutions within each replicate. Tie-break uncertainty and descriptive sampling intervals were reported separately. Filtering improved broad concordance but left leading-list overlap weak. Native boundary scores tied across 412–515 genes.

Finding. Genome-wide agreement does not establish stable short candidate lists. The comparison measures processing sensitivity conditional on available matched specimens; it does not isolate a causal source of the discrepancy.

3. Reference choice, numerical execution and external baselines

Ranking candidates against two normal references

Globally separated case/donor partitions compared adjacent-normal and GTEx references in corrected native and paired study designs. A bootstrap lower-quantile score was evaluated against single-reference effects, mean ranks, reference maximin and the official R `RobustRankAggreg` implementation [2]. Top-25/50/100 lists were evaluated with heldout effect/direction summaries and paired resampling. Bootstrap rank stability, reference ablations and reference discordance were retained as exploratory diagnostics.

The proposed score showed **no top-50 advantage over simple maximin** in the corrected studies. RRA supplies an established ranking baseline; because both input lists share tumor specimens, its independent-list null cannot justify calibrated significance here. A constructed shared-normal-bias example further showed that agreement between references can coexist with incorrect biological selection. These results restrict the interpretation of consensus scores.

Execution can reproduce candidates while failing numerical acceptance

Archived replay compared macOS/Linux with pinned dependencies and Polars 1.42.1/1.44.2. It recorded ordering, tie counts and intermediate arrays, while retaining the original tolerances: $\mathrm{rtol} = 10^{-8}$, $\mathrm{atol} = 10^{-10}$. The three historical failed rows repeated one Spearman statistic at three top-k values. The earliest observed platform divergence was in log2 effect arrays, followed by tie-sensitive ranks; the underlying platform cause was not established by the dependency comparison.

Corrected global studies reproduced candidate lists and heldout metrics, but **63 native and 60 paired sensitivity cells failed** the fixed tolerance. A later exact-arithmetic study compared existing eager rational, specialized rational and CGAL lazy exact implementations: all 46 workload/specification comparisons agreed, including native macOS checks. Of 6,624 process attempts, 36 binary64 attempts failed on a constructed extreme-value fixture; exact-method attempts succeeded. This separate study did not repair the archived numerical failures or establish a new arithmetic algorithm.

Fair external workflow and failure experiments

The pinned `tcga-data-nf` preparation implementation [3] was compared on cached recount3 inputs with matching specimen/gene membership and normalization/filtering semantics. There were 21 comparable preparation workloads and 7 successful native-script integrations. Output equivalence was established before timing. A common R normalization backend, adapter costs and emulated hardware limit performance interpretation; no end-to-end Nextflow or engine-speed advantage was established.

Tested safeguard	Observed behavior	Interpretation
Corrupt input, missing metadata, conflicting duplicates	Benchmark adapter/parser rejected each	Guards worked at the recorded validation layer.
Interrupted execution	No incorrect output committed; fresh retry recovered	Recovery tested with an explicit pause hook.
Stale cached evidence	Evidence hash guard rejected reuse	External native stale-cache handling was unmeasured.
Frozen sources and reports	Hashes, manifests and earlier failures retained	Traceability supports audit; it does not prove biological truth.

Finding. The supported contribution is a measured engineering case study: source-to-claim traceability, controlled comparisons and preservation of failed scientific checks. Passing software tests does not override those failures.

4. Additional method experiments and research status

The project also tested whether its failure cases motivated a sharper method or systems contribution. These exploratory studies used frozen evidence and strong baselines. Their bounded negative results are part of the research record.

Study	Executed evidence	Outcome and boundary
Executable evidence contracts	32 synthetic cases; 192 tool/case/pass decisions using SQL, SHACL and Snakemake	All matched the prototype. Each admitted two false attestations; metadata consistency cannot establish original truth.
Decision-aware recomputation	400 distinct patch events across 40 workloads; 233 unchanged sets certified	No decision mismatches; total cost beat exact incremental updating in 0/40 workloads.
Exact ranking arithmetic	46 comparisons using established exact methods	Exact results agreed; no unique capability or uniformly fastest implementation.
Rank selection with abstention	3,600 synthetic datasets; Holm, Bonferroni and joint bootstrap	Narrow label screen passed; invalid pair directions occurred in 47/200 versus 5/200 datasets in one moderate-signal cell.
Minimum claim-flipping changes	112 synthetic workflow alternatives in 7 scenarios	SQL and Python agreed on all decisions and witnesses; no unique capability demonstrated.

Repeated timings/tool passes are not independent datasets. The abstention diagnostic is the 30-donor moderate-signal cell. Its pairwise error undermines a calibrated-confidence claim despite correct observed top-k labels. The official confident-feature-ranking implementation [4] was not benchmarked. The claim-flipping test replayed prepared synthetic alternatives rather than generating real quantification workflows.

Position relative to prior work

Arora et al. [5] already established that processing differences affect expression and downstream results. CancerOmicsLake adds an auditable combination of matched candidate-effect comparisons, tie diagnostics and explicit numerical failure records; it does not claim to discover processing sensitivity first. Direct Arora replication remains unexecuted: the checked expression-archive locations returned access errors on 16 October 2026. This does not establish that the matrices were never published. Their endpoints and ours differ.

What this research supports, and the next step

The deliverable is a working cancer-omics data platform and a documented engineering/sensitivity case study. It has not established a superior new ranking method, independent biological validation or clinical utility. Batch effects, available-sample selection and annotation differences remain limitations. Existing inspected holdouts cannot provide fresh confirmation.

The next concrete steps are independent investigator reproduction of the prepared handoff, external comparison on identical endpoints, and an untouched evaluation cohort selected before method development. Any future algorithm needs a clear estimand and benefit over maximin or competent systems baselines. For a PhD, my interest is developing and evaluating reliable data infrastructure for biomedical research, with this platform as an implemented starting point.

Document status. Working research summary for interview discussion, updated from the frozen v5 evidence and subsequent feasibility studies. Code and drafting used AI assistance. Author review, affiliation, correspondence and formal declarations remain to be finalized; independent human reproduction is pending. The evidence ledger and reproducible LaTeX build accompany this brief.

References

[1] Wilks et al. *recount3: summaries and queries for large-scale RNA-seq expression and splicing.* Genome Biology, 2021. doi:10.1186/s13059-021-02533-6.

[2] Kolde et al. *Robust rank aggregation for gene list integration and meta-analysis.* Bioinformatics, 2012. doi:10.1093/bioinformatics/btr709.

[3] Quackenbush Lab. *tcga-data-nf.* Pinned workflow source; benchmark scope described on p. 3.

[4] Neuhof and Benjamini. *Confident Feature Ranking.* AISTATS, 2024. PMLR 238:1468–1476.

[5] Arora et al. *Variability in estimated gene expression among commonly used RNA-seq pipelines.* Scientific Reports, 2020. doi:10.1038/s41598-020-59516-z.